Turbo-Structured Bayesian Channel Estimation for Non-Stationary Hybrid-Field RIS-Assisted Systems

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***Abstract*—Reconfigurable intelligent surface (RIS)-assisted millimeter-wave (mmWave) systems with large-scale arrays exhibit hybrid-field propagation and spatial non-stationarity, which render conventional far-field and fully visible array-based channel models inadequate and make channel estimation challenging. In this paper, we investigate the non-stationary cascaded channel estimation problem in a hybrid-field scenario, where the RIS-base station (BS) link operates in the far-field and the user-RIS link exhibits near-field characteristics with partially visible RIS elements, referred to as visible regions (VRs). By exploiting the structural characteristics of the cascaded channel, a reduced-dimensional sparse bilinear representation is constructed via a visibility-weighted polar-domain dictionary compression strategy. Based on this representation, a turbo-structured joint Bayesian estimation (TS-JBE) approach is proposed to jointly estimate the channel gains, VRs, and off-grid parameters. Simulation results demonstrate that the proposed method significantly improves the estimation accuracy compared with existing approaches.**

***Index Terms*—reconfigurable intelligent surface, hybrid-field, channel estimation, visible region, Bayesian estimation.**

I. Introduction

With the rapid growth of mobile devices and wireless applications, future wireless networks face challenges in peak data rate, spectral efficiency, energy efficiency, and reliability [1], [2]. As a key enabling technology for sixth-generation (6G) networks, reconfigurable intelligent surfaces (RISs) can reconfigure the wireless propagation environment in a cost-effective and energy-efficient manner, providing new degrees of freedom for system performance [3]–[5]. However, due to the passive nature of RIS elements and the high dimension of cascaded channels in large-scale RIS deployments, acquiring channel state information (CSI) is challenging [6]. Although an on-off reflection scheme can decouple the cascaded channels [7], it incurs high pilot overhead. To mitigate this, methods under fully reflective operation based on Least Squares (LS) / Linear Minimum Mean Square Error (LMMSE) have been proposed [8], [9], along with compressed sensing and matrix factorization techniques exploiting channel sparsity [10]–[13].

The aforementioned works assume far-field planar-wave propagation. However, for large-scale RISs at millimeter-wave (mmWave) or terahertz frequencies, users or scatterers often lie in the near-field of the RIS, requiring spherical-wave models. In this context, near-field channels exhibit sparsity in the polar domain [14]. Motivated by this, hybrid-field cascaded channel models and estimation methods have been systematically studied [15]–[17]. Moreover, large-scale RISs introduce spatial non-stationarity, where different scattering paths illuminate only partial regions, referred to as visible regions (VRs) [18], and ignoring this property leads to model mismatch. To address this issue, joint estimation of VRs and channels has been investigated [19], [20], but these methods are limited by either incomplete multipath modeling or error propagation.

Motivated by the above observations, this paper investigates the cascaded channel estimation problem in non-stationary hybrid-field RIS-assisted communication systems. Specifically, we propose a turbo-structured joint Bayesian estimation (TS-JBE) approach, which exploits structured channel priors to jointly estimate cascaded channel gains, VRs, and off-grid parameters, thereby effectively capturing channel characteristics and mitigating error propagation in stage-wise approaches. Simulation results show that TS-JBE achieves superior normalized mean squared error (NMSE) performance compared with existing schemes, validating its effectiveness.

II. System Model

As illustrated in Fig. 1, consider a RIS-aided communication system operating in the mmWave band under the time-division duplex (TDD) mode. The base station (BS) is equipped with M antennas, while the RIS consists of N reflecting elements with $N \gg M$. A single-antenna user is served, with the RIS deployed in its vicinity to enhance the communication performance.[1] Both the BS and RIS employ uniform linear arrays (ULAs) with half-wavelength spacing, i.e., $d = \lambda/2$. The user-BS link is assumed to be blocked, and thus communication relies solely on the cascaded user-RIS and RIS-BS channels, with $\mathbf{h}_u \in \mathbb{C}^{N\times 1}$ and $\mathbf{H} \in \mathbb{C}^{M\times N}$ denoting the corresponding channels, respectively.

[1]The proposed approach can be extended to multiuser scenarios by assigning orthogonal pilot sequences to different users.

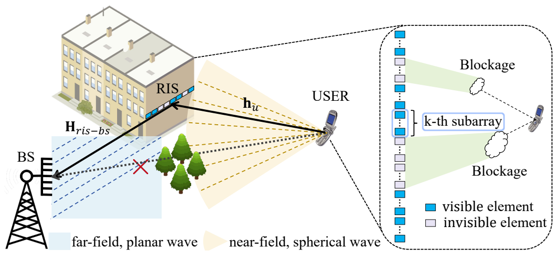


Fig. 1. Considered RIS-aided mmWave system model.

A. Channel Model

For accurate channel modeling, it is necessary to distinguish different propagation regions. Specifically, the near- and far-field regions are separated by the Rayleigh distance $Z = \frac{2(D_r^2+D_s^2)}{\lambda}$, where D_r and D_s denote the receive and transmit array apertures, and λ is the carrier wavelength [1]. In the considered system, the RIS is deployed near the user, rendering the user-RIS link near-field, while the RIS-BS link remains far-field [21]. Accordingly, spherical- and planar-wave models are applied to the user-RIS and RIS-BS links, respectively. For the user-RIS channel, the near-field array response vector (ARV) is derived under the Fresnel approximation [22]. Specifically, the ARV for the l-th path is given by $\mathbf{a}(\vartheta_{U,l}, r_{U,l}) = [e^{-j\frac{2\pi}{\lambda}(r_{U,l}^{(1)}-r_{U,l})}, \ldots, e^{-j\frac{2\pi}{\lambda}(r_{U,l}^{(N)}-r_{U,l})}]^T$, where $\vartheta_{U,l} = \cos\theta_{U,l}$ represents the spatial angle, $r_{U,l}$ is the distance to the reference RIS element, and $r_{U,l}^{(n)}$ denotes the distance from the user/scatterer to the n-th RIS element, which is approximated as $r_{U,l} - (n-1)d\,\vartheta_{U,l} + \frac{d^2(n-1)^2(1-\vartheta_{U,l}^2)}{2r_{U,l}}$. To capture spatial non-stationarity, a binary visibility vector $\boldsymbol{\phi}_l \in \{0,1\}^N$ is introduced for the l-th path [23]–[25], where $\phi_{l,n} = 1$ indicates visibility of the n-th RIS element to the l-th path. With this definition, the user-RIS channel is modeled as

$$\mathbf{h}_u = \sum_{l=1}^{L_U} \alpha_l\, \mathbf{a}(\vartheta_{U,l}, r_{U,l}) \odot \boldsymbol{\phi}_l, \tag{1}$$

where α_l denotes the complex gain and L_U denotes the number of user-RIS propagation paths. For the RIS-BS link, the l-th path is described by the BS angle of arrival (AoA) $\vartheta_{B,l} = \cos\theta_{B,l}$ and the RIS angle of departure (AoD) $\varphi_{R,l} = \cos\phi_{R,l}$, with the corresponding far-field ARVs given by $\mathbf{a}_B(\vartheta_{B,l}) = [1, e^{-j\frac{2\pi}{\lambda}d\vartheta_{B,l}}, \ldots, e^{-j\frac{2\pi}{\lambda}(M-1)d\vartheta_{B,l}}]^T$ and $\mathbf{a}_R(\varphi_{R,l}) = [1, e^{-j\frac{2\pi}{\lambda}d\varphi_{R,l}}, \ldots, e^{-j\frac{2\pi}{\lambda}(N-1)d\varphi_{R,l}}]^T$. Accordingly, the RIS-BS channel can be expressed as

$$\mathbf{H} = \sum_{l=1}^{L_{RB}} \beta_l\, \mathbf{a}_B(\vartheta_{B,l})\mathbf{a}_R^H(\varphi_{R,l}), \tag{2}$$

where β_l denotes the complex gain and L_{RB} denotes the number of RIS-BS propagation paths. Moreover, define $\boldsymbol{\Theta} = \mathrm{diag}(\boldsymbol{\eta})$ as the RIS reflection matrix, where $\boldsymbol{\eta} = [e^{j\nu_1}, \ldots, e^{j\nu_N}]^T$, and $\nu_n \in [0, 2\pi)$ denotes the phase shift of the n-th element. Accordingly, the cascaded channel is given by $\mathbf{h} = \mathbf{H}\boldsymbol{\Theta}\mathbf{h}_u$. In addition, during uplink training, the user transmits P pilot symbols with transmit power P_t. After pilot correlation and normalization, the equivalent observation is

$$\mathbf{y}_p = \mathbf{H}\boldsymbol{\Theta}_p\mathbf{h}_u + \mathbf{n}_p = \mathbf{h}_p + \mathbf{n}_p, \quad \mathbf{n}_p \sim \mathcal{CN}(\mathbf{0}, \tfrac{\sigma^2}{P_t}\mathbf{I}_M), \tag{3}$$

where $\mathbf{h}_p \triangleq \mathbf{H}\boldsymbol{\Theta}_p\mathbf{h}_u$ denotes the effective cascaded channel associated with the p-th pilot and σ^2 denotes the noise variance. Stacking the signals over all pilot intervals yields

$$\tilde{\mathbf{y}} = [\mathbf{y}_1^T, \ldots, \mathbf{y}_P^T]^T = \tilde{\mathbf{h}} + \tilde{\mathbf{n}}, \tag{4}$$

where $\tilde{\mathbf{h}} = [\mathbf{h}_1^T, \ldots, \mathbf{h}_P^T]^T$ and $\tilde{\mathbf{n}} = [\mathbf{n}_1^T, \ldots, \mathbf{n}_P^T]^T$.

III. Sparse Channel Representation and Prior Modeling

In this section, a sparse representation of the cascaded channel is developed, and the associated prior models are introduced by exploiting the structural characteristics of the RIS-aided hybrid-field propagation environment. For clarity, a generic pilot p is considered, and the index is omitted whenever no ambiguity arises. The received signal, cascaded channel, and noise are denoted by $\mathbf{y}$, $\mathbf{h}$, and $\mathbf{n}$, respectively.

A. Sparse Representation and Model Transformation

1) Sparse Representation: To exploit the sparsity of the cascaded channel, uniform angular grids $\boldsymbol{\vartheta} \in \mathbb{R}^{M\times 1}$ and $\boldsymbol{\varphi} \in \mathbb{R}^{N\times 1}$ are employed at the BS and RIS, yielding the angular-domain dictionaries $\mathbf{F}_M \in \mathbb{C}^{M\times M}$ and $\mathbf{F}_N \in \mathbb{C}^{N\times N}$. Accordingly, the RIS-BS channel admits the sparse representation $\mathring{\mathbf{H}} = \mathbf{F}_M\mathbf{A}\mathbf{F}_N^H$, where $\mathbf{A}$ is a sparse gain matrix and the superscript $(\circ)$ denotes the ideal on-grid case. For the near-field user-RIS link, a polar-domain sparse representation with a non-uniform grid $[\boldsymbol{\vartheta}_u, \mathbf{r}_u] \in \mathbb{R}^{\bar{N}\times 2}$ is adopted [14], i.e., $\mathring{\mathbf{h}}_u = (\mathbf{W} \odot \boldsymbol{\Phi})\mathbf{b}$, where $\mathbf{W} \in \mathbb{C}^{N\times\bar{N}}$ is the polar-domain dictionary, $\boldsymbol{\Phi}$ is the binary VR matrix, $\mathbf{b}$ is the sparse vector, and $\odot$ denotes the Hadamard product. Consequently, the overall hybrid-field channel can be expressed as

$$\mathring{\mathbf{h}} = \mathbf{F}_M\mathbf{A}\mathbf{F}_N^H\mathrm{diag}(\boldsymbol{\eta})(\mathbf{W} \odot \boldsymbol{\Phi})\mathbf{b}. \tag{5}$$

2) Model Transformation: The cascaded channel in (5) exhibits a multilinear coupling among $\mathbf{A}$, $\boldsymbol{\Phi}$, and $\mathbf{b}$. Using vectorization and Khatri-Rao identities [26], it admits the form

$$\mathring{\mathbf{h}} = (\boldsymbol{\eta}^T \otimes \mathbf{I}_M)((\bar{\mathbf{W}} * \mathbf{F}_N^*) \otimes \mathbf{F}_M)\mathrm{vec}(\mathbf{b}^T \otimes \mathbf{A}), \tag{6}$$

where $\bar{\mathbf{W}} = \mathbf{W} \odot \boldsymbol{\Phi}$, and $\otimes$ and $*$ denote the Kronecker product and the row-wise Khatri-Rao product, respectively. Direct estimation of $\mathrm{vec}(\mathbf{b}^T \otimes \mathbf{A}) \in \mathbb{C}^{MN\bar{N}\times 1}$ is prohibitive due to the high dimension of the coupled dictionaries. Based on [20, Lemma 1], it can be further shown that $\bar{\mathbf{W}} * \mathbf{F}_N^* \Leftrightarrow \mathbf{Q} \odot \bar{\boldsymbol{\Phi}}$, where $\mathbf{Q} \in \mathbb{C}^{N\times\bar{Q}}$ is the equivalent near-field dictionary with grid $[\bar{\varphi}, \bar{r}] \in \mathbb{R}^{\bar{Q}\times 2}$, $\bar{\boldsymbol{\Phi}} \in \{0,1\}^{N\times\bar{Q}}$ is the equivalent VR matrix, and $\Leftrightarrow$ indicates that the transformation preserves the number of active columns in $\mathring{\mathbf{h}}$. Therefore, the cascaded channel can be rewritten as

$$\mathring{\mathbf{h}} = (\boldsymbol{\eta}^T \otimes \mathbf{I}_M)((\mathbf{Q} \odot \bar{\boldsymbol{\Phi}}) \otimes \mathbf{F}_M)\,\mathrm{vec}(\mathbf{X}), \tag{7}$$

where $\mathbf{X} \in \mathbb{C}^{M\times\bar{Q}}$ is the joint sparse matrix corresponding to the transformed dictionary $(\mathbf{Q}\odot\bar{\boldsymbol{\Phi}})\otimes\mathbf{F}_M$. This transformation replaces the coupled hybrid-field visibility-weighted dictionary with a single near-field visibility-weighted dictionary, reducing the sparse vector dimension from $\mathcal{O}(MN^2)$ to $\mathcal{O}(MN)$, since $\bar{Q}$ and N share the same grid generation algorithm for the same array. Based on the above representation and further matrix transformation, the cascaded channel is given by

$$\overset{\circ}{\mathbf{h}} = \mathbf{D}_1(\bar{\mathbf{\Phi}})\,\boldsymbol{x} = \mathbf{D}_2(\boldsymbol{x})\,\mathrm{vec}(\bar{\mathbf{\Phi}}), \tag{8}$$

where $\boldsymbol{x} \triangleq \mathrm{vec}(\mathbf{X})$, $\mathbf{D}_1(\bar{\mathbf{\Phi}}) = (\boldsymbol{\eta}^T \otimes \mathbf{I}_M)\,((\mathbf{Q} \odot \bar{\mathbf{\Phi}}) \otimes \mathbf{F}_M)$, $\mathbf{D}_2(\boldsymbol{x}) = (\boldsymbol{x}^T \otimes (\boldsymbol{\eta}^T \otimes \mathbf{I}_M))\,\mathrm{diag}(\mathrm{vec}(\mathbf{Q} \otimes \mathbf{F}_M))\,\mathbf{S}(\mathbf{1}_{M\times M})$ with $\mathbf{S}(\mathbf{1}_{M\times M})$ computed according to Lemma 1. This implies that $\overset{\circ}{\mathbf{h}}$ admits a linear representation with respect to both the cascaded channel gain vector $\boldsymbol{x}$ and the VR matrix $\bar{\mathbf{\Phi}}$.

Lemma 1. *For any matrices* $\mathbf{A} = [\mathbf{a}_1, ..., \mathbf{a}_{c_A}] \in \mathbb{C}^{r_A \times c_A}$ *and* $\mathbf{B} = [\mathbf{b}_1, ..., \mathbf{b}_{c_B}] \in \mathbb{C}^{r_B \times c_B}$*, there exists a transformation matrix* $\mathbf{S}(\mathbf{A}) \in \mathbb{C}^{r_B c_B r_A c_A \times r_B c_B}$ *such that*

$$\mathrm{vec}(\mathbf{B} \otimes \mathbf{A}) = \mathbf{S}(\mathbf{A})\,\mathrm{vec}(\mathbf{B}), \tag{9}$$

where $\mathbf{S}(\mathbf{A})$ *is explicitly given by* $\mathbf{S}(\mathbf{A}) = \mathbf{I}_{c_B} \otimes \begin{bmatrix} \mathbf{I}_{r_B} \otimes \mathbf{a}_1 \\ \vdots \\ \mathbf{I}_{r_B} \otimes \mathbf{a}_{c_A} \end{bmatrix}$.

Proof. The proof is omitted for brevity. □

3) Off-Grid Modeling: To mitigate basis mismatch caused by continuous-valued angles and distances, off-grid parameters $\mathbf{\Xi} = \{\Delta\bar{\boldsymbol{\varphi}}, \Delta\bar{\mathbf{r}}, \Delta\boldsymbol{\vartheta}\}$ are introduced [16]. With the off-grid set $\mathbf{\Xi}$, the practical cascaded channel is modeled as

$$\mathbf{h} = \mathbf{D}_1(\bar{\mathbf{\Phi}}, \mathbf{\Xi})\,\boldsymbol{x} = \mathbf{D}_2(\boldsymbol{x}, \mathbf{\Xi})\,\mathrm{vec}(\bar{\mathbf{\Phi}}), \tag{10}$$

where $\mathbf{D}_1(\bar{\mathbf{\Phi}}, \mathbf{\Xi}) = (\boldsymbol{\eta}^T \otimes \mathbf{I}_M)\,((\mathbf{Q}(\Delta\bar{\boldsymbol{\varphi}}, \Delta\bar{\mathbf{r}}) \odot \bar{\mathbf{\Phi}}) \otimes \mathbf{F}_M(\Delta\boldsymbol{\vartheta}))$ and $\mathbf{D}_2(\boldsymbol{x}, \mathbf{\Xi}) = (\boldsymbol{x}^T \otimes (\boldsymbol{\eta}^T \otimes \mathbf{I}_M))\,\mathrm{diag}(\mathrm{vec}(\mathbf{Q}(\Delta\bar{\boldsymbol{\varphi}}, \Delta\bar{\mathbf{r}}) \otimes \mathbf{F}_M(\Delta\boldsymbol{\vartheta})))\,\mathbf{S}(\mathbf{1}_{M\times M})$.

B. Hierarchical Sparse Prior Model

To exploit the sparsity structure of $\boldsymbol{x}$, a structure-aware three-layer hierarchical sparse (3LHS) prior [27] is adopted:

$$p(\boldsymbol{x}, \boldsymbol{\rho}, \boldsymbol{s}) = p(\boldsymbol{s})\,p(\boldsymbol{\rho}|\boldsymbol{s})\,p(\boldsymbol{x}|\boldsymbol{\rho}), \tag{11}$$

where $\boldsymbol{s}$ is an independent Bernoulli binary vector, with each element indicating whether the corresponding entry of $\boldsymbol{x}$ is nonzero ($s_{m,q} = 1$ if $x_{m,q} \neq 0$, and 0 otherwise), and $p(\boldsymbol{s}) = \prod_{m,q} \lambda_{m,q}^{s_{m,q}} (1-\lambda_{m,q})^{1-s_{m,q}}$, where $\lambda_{m,q}$ characterizes the sparsity of $\boldsymbol{x}$. Conditioned on $\boldsymbol{s}$, $\boldsymbol{\rho}$ follows $p(\boldsymbol{\rho}|\boldsymbol{s}) = \prod_{m,q} \Gamma(\rho_{m,q}; a_{m,q}, b_{m,q})^{s_{m,q}} \Gamma(\rho_{m,q}; \bar{a}_{m,q}, \bar{b}_{m,q})^{1-s_{m,q}}$, where $\Gamma(\rho; \cdot, \cdot)$ denotes a Gamma distribution with a pair of hyperparameters, used to control the variance of $\boldsymbol{x}$ via a, b and $\bar{a}, \bar{b}$. Conditioned on $\boldsymbol{\rho}$, each entry of $\boldsymbol{x}$ is modeled as a zero-mean complex Gaussian $p(\boldsymbol{x}|\boldsymbol{\rho}) = \prod_{m,q} \mathcal{CN}(x_{m,q}; 0, \rho_{m,q}^{-1})$.

C. Markov Clustering Prior Model

As in Section II-A, the visibility of RIS elements is represented by 0/1 variables, which tend to form clusters due to spatial correlation [28]. To exploit this, the RIS is divided into K subarrays, with all elements in a subarray sharing the same visibility. Let $v_{k,q}$ denote the visibility of the k-th subarray at the q-th grid point. Accordingly, the full RIS VR matrix can be expressed as $\bar{\mathbf{\Phi}} = \bar{\mathbf{\Phi}}_s \otimes \mathbf{1}_{\frac{N}{K}\times 1}$, where $\bar{\mathbf{\Phi}}_s \in \{0, 1\}^{K \times \bar{Q}}$ denotes the subarray VR matrix. Moreover, inter-subarray correlation is captured by a Markov chain along subarrays [25]: $p(\bar{\mathbf{\Phi}}_s) = \prod_{q=1}^{\bar{Q}} p(v_{1,q}) \prod_{k=2}^{K} p(v_{k,q}|v_{k-1,q})$, with initial and transition probabilities given by $p(v_{1,q} = 1) = \lambda_{\mathrm{VR}}$, $p(v_{k,q} = 1|v_{k-1,q} = 0) = p_{01}$, $p(v_{k,q} = 0|v_{k-1,q} = 1) = p_{10}$, where $\lambda_{\mathrm{VR}} = \frac{p_{01}}{p_{01}+p_{10}}$ denotes the visibility ratio of RIS elements.

D. Joint Probabilistic Model of the Received Signal

As shown in (4), the received signal $\tilde{\mathbf{y}}$ follows a linear observation model with complex Gaussian noise $\tilde{\mathbf{n}} \sim \mathcal{CN}(\mathbf{0}, \kappa^{-1}\mathbf{I})$, where the noise precision κ has a Gamma prior, $p(\kappa) = \Gamma(\kappa; c, d)$. Accordingly, the joint distribution over $\tilde{\mathbf{y}}, \boldsymbol{x}, \bar{\mathbf{\Phi}}_s, \kappa, \boldsymbol{\rho}, \boldsymbol{s}$, and $\mathbf{\Xi}$ can be factorized as

$$\begin{aligned} p(\tilde{\mathbf{y}}, \boldsymbol{x}, \bar{\mathbf{\Phi}}_s, \boldsymbol{\rho}, \boldsymbol{s}, \kappa; \mathbf{\Xi}) &= p(\tilde{\mathbf{y}}|\boldsymbol{x}, \bar{\mathbf{\Phi}}_s, \kappa; \mathbf{\Xi}) \\ &\quad \times p(\boldsymbol{x}, \boldsymbol{\rho}, \boldsymbol{s})\,p(\bar{\mathbf{\Phi}}_s)\,p(\kappa), \end{aligned} \tag{12}$$

where $p(\tilde{\mathbf{y}}|\boldsymbol{x}, \bar{\mathbf{\Phi}}_s, \kappa; \mathbf{\Xi})$ denotes the likelihood function of $\tilde{\mathbf{y}}$. Specifically, it follows a complex Gaussian distribution given by $\mathcal{CN}(\tilde{\mathbf{D}}_1(\bar{\mathbf{\Phi}}_s, \mathbf{\Xi})\boldsymbol{x}, \kappa^{-1}\mathbf{I})$, where $\tilde{\mathbf{D}}_1(\bar{\mathbf{\Phi}}_s, \mathbf{\Xi})$ is constructed by row-wise concatenation of the sensing matrices corresponding to the P pilot symbols. In addition, $p(\boldsymbol{x}, \boldsymbol{\rho}, \boldsymbol{s})$ and $p(\bar{\mathbf{\Phi}}_s)$ are given in Section III-B and III-C. Building upon the above probabilistic model, a Bayesian inference framework is adopted. Within this framework, the TS-JBE algorithm is proposed, which exploits the structural characteristics of the channel to jointly estimate the cascaded channel parameters.

IV. Turbo-Structured Joint Bayesian Estimation

A. Outline of the TS-JBE Algorithm

The TS-JBE algorithm alternates among three modules, as shown in Fig. 2, with their functions summarized below.

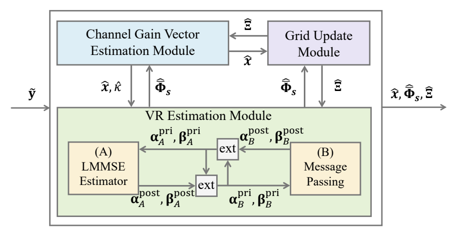


Fig. 2. Flow chart of the proposed algorithm.

- **Channel Gain Estimation Module:** Given $\tilde{\mathbf{y}}$, $\hat{\bar{\mathbf{\Phi}}}_s$, and $\hat{\mathbf{\Xi}}$, the cascaded channel gain vector $\boldsymbol{x}$ and noise precision κ are estimated by optimizing the posterior distribution.
- **VR Estimation Module:** Given $\tilde{\mathbf{y}}$, $\hat{\boldsymbol{x}}$, $\hat{\mathbf{\Xi}}$, and $\hat{\kappa}$, the subarray VR matrix $\bar{\mathbf{\Phi}}_s$ is updated via the LMMSE estimator combined with message passing (MP).
- **Grid Update Module:** The off-grid parameters $\mathbf{\Xi}$ are refined via gradient descent on the residual between $\tilde{\mathbf{y}}$ and its reconstruction from $\hat{\boldsymbol{x}}$ and $\hat{\bar{\mathbf{\Phi}}}_s$.

B. Channel Gain Vector Estimation Module

Given $\tilde{\mathbf{y}}$, $\hat{\bar{\mathbf{\Phi}}}_s$, and $\hat{\mathbf{\Xi}}$, the cascaded channel gain vector $\boldsymbol{x}$ and the noise precision κ are inferred via posterior optimization to obtain maximum a posteriori (MAP) estimates, i.e., $\boldsymbol{x}^* = \arg\max_{\boldsymbol{x}} p(\boldsymbol{x}|\tilde{\mathbf{y}}, \hat{\bar{\mathbf{\Phi}}}_s; \hat{\mathbf{\Xi}})$ and $\kappa^* = \arg\max_{\kappa} p(\kappa|\tilde{\mathbf{y}}, \hat{\bar{\mathbf{\Phi}}}_s; \hat{\mathbf{\Xi}})$. Based on (12), the joint estimation of $\boldsymbol{x}$ and κ is a high-dimensional Bayesian problem with coupled latent variables. To address this issue, a structured-subspace-constrained variational Bayesian inference (SSC-VBI) method is proposed. Define $\mathbf{\Psi} \triangleq \{\boldsymbol{x}, \boldsymbol{\rho}, \boldsymbol{s}, \kappa\}$. Let ψ_j, $j \in \mathcal{H}$, denote the j-th element of $\mathbf{\Psi}$, where $\mathcal{H}$ is the corresponding index set. Denote the posterior of $\mathbf{\Psi}$ by $p(\mathbf{\Psi} \mid \tilde{\mathbf{y}})$. Following the VBI method, the approximate posterior $q(\mathbf{\Psi})$ is obtained by solving [29]

$$\mathcal{A}_{\text{VBI}}: \quad \min_{q(\boldsymbol{\Psi})} \int q(\boldsymbol{\Psi}) \ln \frac{q(\boldsymbol{\Psi})}{p(\boldsymbol{\Psi}|\tilde{\mathbf{y}})} \, d\boldsymbol{\Psi}$$
$$\text{s.t. } q(\boldsymbol{\Psi}) = \prod_{j \in \mathcal{H}} q(\psi_j), \int q(\psi_j)\, d\psi_j = 1, \quad (13)$$
$$q(\boldsymbol{x}) = \mathcal{CN}(\boldsymbol{x}; \boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2)).$$

Therefore, we propose SSC-VBI, which builds upon subspace-constrained VBI (SC-VBI) [30] to solve the above problem. Specifically, we optimize the subspace selection strategy by adopting column-wise selection for $\mathbf{X}$, selecting at most one active index per column, since each column contains at most one nonzero element corresponding to a single BS-side AoA. Except for this selection strategy, the specific formulas are the same as those of SC-VBI and are therefore omitted.

C. VR Estimation Module

Given $\tilde{\mathbf{y}}$, $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{\Xi}}$ and $\hat{\kappa}$, the subarray VR matrix $\bar{\boldsymbol{\Phi}}_s$ is also estimated under the MAP criterion. Before estimation, the observation model is simplified by leveraging the sparsity of $\hat{\boldsymbol{x}}$ and the real-valued nature of the VR. Let $\mathcal{Q}_v$ denote the active column set of $\bar{\boldsymbol{\Phi}}_s$ (obtained via thresholding $\hat{\boldsymbol{x}}$). Estimation is then restricted to the corresponding submatrix $\bar{\boldsymbol{\Phi}}_{s,\mathcal{Q}_v} = \bar{\boldsymbol{\Phi}}_s(:, \mathcal{Q}_v) \in \mathbb{R}^{K \times |\mathcal{Q}_v|}$ with corresponding measurement matrix $\tilde{\mathbf{D}}_{2,\mathcal{Q}_v}$, preserving the dominant paths while greatly reducing complexity. Moreover, since $\bar{\boldsymbol{\Phi}}_{s,\mathcal{Q}_v}$ is real-valued, the model is converted to an equivalent real-valued form $\bar{\mathbf{y}} = [\Re(\tilde{\mathbf{y}})^T, \Im(\tilde{\mathbf{y}})^T]^T$, and define $\bar{\mathbf{D}}_{2,\mathcal{Q}_v}$ and $\bar{\mathbf{n}}$ correspondingly. Then, the real-valued model is given by

$$\bar{\mathbf{y}} = \bar{\mathbf{D}}_{2,\mathcal{Q}_v} \mathrm{vec}(\bar{\boldsymbol{\Phi}}_{s,\mathcal{Q}_v}) + \bar{\mathbf{n}}. \quad (14)$$

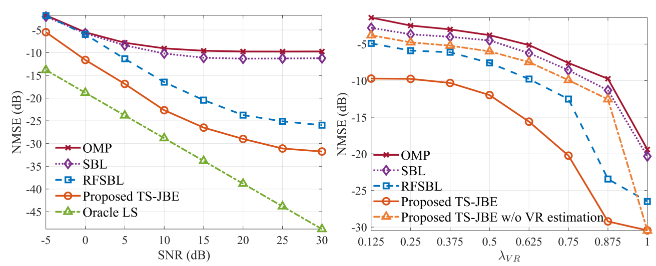
With this reformulation, we introduce a turbo framework consisting of an LMMSE module and a message passing (MP) module. The framework estimates parameters via iterative exchanges, leveraging the clustering prior, as shown in Fig. 2. Specifically, Module A exploits the prior information $(\boldsymbol{\alpha}_{A,\text{pri}}, \boldsymbol{\beta}_{A,\text{pri}})$ to construct a Gaussian prior for $\bar{\boldsymbol{\Phi}}_{s,\mathcal{Q}_v}$ and performs LMMSE estimation to obtain $(\boldsymbol{\alpha}_{A,\text{post}}, \boldsymbol{\beta}_{A,\text{post}})$. Module B adopts Markov chain MP [27] to capture the clustering property of VRs. Moreover, Module ext is used to generate prior information by decorrelating the noise between the input and output signals. The iterative exchanges between these modules continue until convergence, after which hard decision and dimension recovery are performed to obtain $\hat{\bar{\boldsymbol{\Phi}}}_s$.

D. Grid Update Module

With $\hat{\bar{\boldsymbol{\Phi}}}_s$ and $\hat{\boldsymbol{x}}$ fixed, the off-grid parameters $\boldsymbol{\Xi}$ are sequentially refined by maximizing the likelihood $L(\Delta\bar{\boldsymbol{\varphi}}, \Delta\bar{\mathbf{r}}, \Delta\boldsymbol{\vartheta}) = -\sum_{p=1}^{P} \|\mathbf{y}_p - (\boldsymbol{\eta}_p^T \otimes \mathbf{I}_M)((\mathbf{Q}(\Delta\bar{\boldsymbol{\varphi}}, \Delta\bar{\mathbf{r}}) \odot \hat{\bar{\boldsymbol{\Phi}}}) \otimes \mathbf{F}_M(\Delta\boldsymbol{\vartheta}))\, \hat{\boldsymbol{x}}\|^2$, using gradient descent with step sizes determined via backtracking line search [31]. Overall, the computational complexity of TS-JBE is dominated by the LMMSE module in VR estimation and the VBI updates in channel gain estimation.

V. Simulation Results

Numerical results validate the proposed TS-JBE algorithm in a RIS-assisted communication system at $f = 28$ GHz under TDD mode. The BS and RIS employ ULAs with $M = 16$ and $N = 128$ elements, respectively, while the user is equipped with a single antenna. The BS and RIS are aligned along the y- and x-axes with reference positions $[-90\text{m}, -30\text{m}]^T$ and $[0\text{m}, 0\text{m}]^T$, respectively. The user-RIS distance is between 10 and 20 m. The large-scale path loss follows [32]. Unless stated otherwise, $L_{RB} = L_U = 3$ paths are considered, the polar-domain dictionary has $\bar{Q} = 256$ points ($Q_p = 64$ angular, $S = 4$ radial), the RIS is split into $K = 8$ subarrays, the VR visibility ratio is 0.875, and SNR is 20 dB. The NMSE of the cascaded channel $\mathbf{h}_c = \mathrm{vec}(\mathbf{H}\mathrm{diag}(\mathbf{h}_u))$ is adopted as the performance metric, $\mathrm{NMSE} = \frac{\|\hat{\mathbf{h}}_c - \mathbf{h}_c\|^2}{\|\mathbf{h}_c\|^2}$, where the reconstructed channel is obtained as $\hat{\mathbf{h}}_c = ((\mathbf{Q}(\Delta\bar{\boldsymbol{\varphi}}, \Delta\bar{\mathbf{r}}) \odot \hat{\bar{\boldsymbol{\Phi}}}) \otimes \mathbf{F}_M(\Delta\boldsymbol{\vartheta}))\hat{\boldsymbol{x}}$. The proposed method is compared with the following benchmarks: *OMP*; *sparse Bayesian learning (SBL)*; *robust fast SBL (RFSBL)* [20]; *TS-JBE without VR estimation*; and *Oracle LS* method with perfect knowledge. TS-JBE converges to a steady NMSE in about 25 iterations. Fig. 3 presents the simulation results. Fig. 3(a) shows that TS-JBE consistently outperforms



(a) (b)

Fig. 3. NMSE performance of the considered algorithms versus (a) SNR and (b) VR visibility ratio.

OMP, SBL, and RFSBL over all SNRs. While OMP and SBL saturate due to ignoring non-stationarity and off-grid effects, and RFSBL is limited by error propagation in sequential estimation, TS-JBE achieves stable gains by leveraging the channel's structured sparsity and performing joint estimation, maintaining a nearly constant gap to Oracle LS at low-to-medium SNRs. Fig. 3(b) shows the NMSE versus λ_{VR}. As λ_{VR} increases, all algorithms benefit from more visible RIS elements. VR-aware schemes outperform OMP and SBL, highlighting the importance of VR modeling in non-stationary channels. Moreover, TS-JBE consistently outperforms RFSBL, even under low visibility, showing stable performance.

VI. Conclusion

This paper investigates the non-stationary cascaded channel estimation problem in RIS-assisted mmWave systems under a hybrid-field propagation environment. By leveraging the structural characteristics of the cascaded channel, a visibility-aware low-dimensional polar-domain representation is developed, based on which a TS-JBE approach is proposed to jointly estimate the channel gains, VRs, and off-grid parameters. Simulation results show that the proposed method consistently outperforms existing schemes in terms of NMSE performance.

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